SALMONELLA-HELA DUAL RNA-SEQ

Master Runbook

PRJNA280839 / GSE67758 — The Founding Dual RNA-seq Study

Project	PRJNA280839 Salmonella-HeLa Dual RNA-seq (GSE67758 subset)
Study	Westermann et al. 2016, Nature 529(7587):496-501 (PMID 26789254)
Environment	Slurm HPC (Task 1) + R/Bioconductor (Task 2)
Curriculum	MIBO8110 Applied Omics
Document type	Operational Standard Operating Procedure (SOP) Runbook
Companion document	README.md (full narrative, data provenance, biological interpretation)

Generated for internal course and reproducibility use.

0. Document Control

Field	Value
Scope	Task 1 (raw reads → counts, HPC bash, dual RNA-seq) and Task 2 (DESeq2 + ggplot2, R)
Audience	Students and instructors running this exercise
Companion doc	README.md (study background, data provenance, known limitations)
Verification status	All accessions/references independently verified against NCBI/GEO/SRA/Ensembl/CrossRef. Scripts reviewed line-by-line and syntax-checked, not executed end-to-end.

1. Purpose & Scope

This Runbook is the step-by-step operational companion to README.md. It assumes a Slurm HPC account for Task 1 and an R/RStudio environment with Bioconductor access for Task 2.

The project: the founding dual RNA-seq study (Westermann et al. 2016), which established the concatenated-reference method this curriculum's companion PRJNA1091619 pipeline also uses. An HPC pipeline reproduces gene-level counts from raw reads; a full DESeq2 + ggplot2 analysis runs on the actual published combined host+pathogen count matrix and independently recomputes the paper's headline finding.

> **The source list's accession was wrong**
>
> GSE73225 (as given) is an unrelated mouse heart-valve study. The real accession was found independently: PubMed → linked GEO records → GSE60144 (SuperSeries) → GSE67758 (the relevant sub-series). See README for the full correction.

2. Study Background

Field	Value
Study	Westermann AJ, et al. (2016) Dual RNA-seq unveils noncoding RNA functions in host-pathogen interactions. Nature 529(7587):496-501.
DOI	10.1038/nature16547
PMID	26789254
GEO SuperSeries	GSE60144
GEO sub-series used	GSE67758 (51 samples total; this project uses 8)
BioProject	PRJNA280839 (SRA study SRP057065)
Host	Homo sapiens, HeLa-S3 cell line
Pathogen	Salmonella enterica serovar Typhimurium str. SL1344
Headline finding	pinT (locus_tag ncRNA0164): sRNA induced during infection, represses invasion genes once intracellular

3. The 8-Sample Subset

GSM	Condition	Timepoint	Replicate	SRR
GSM1924778	infected (WT)	4h	1	SRR2936892
GSM1924779	infected (WT)	4h	2	SRR2936893
GSM1924787	infected (WT)	24h	1	SRR2936901
GSM1924788	infected (WT)	24h	2	SRR2936902
GSM1924793	mock	4h	1	SRR2936907
GSM1924794	mock	4h	2	SRR2936908
GSM1924802	mock	24h	1	SRR2936916
GSM1924803	mock	24h	2	SRR2936917

4. Prerequisites

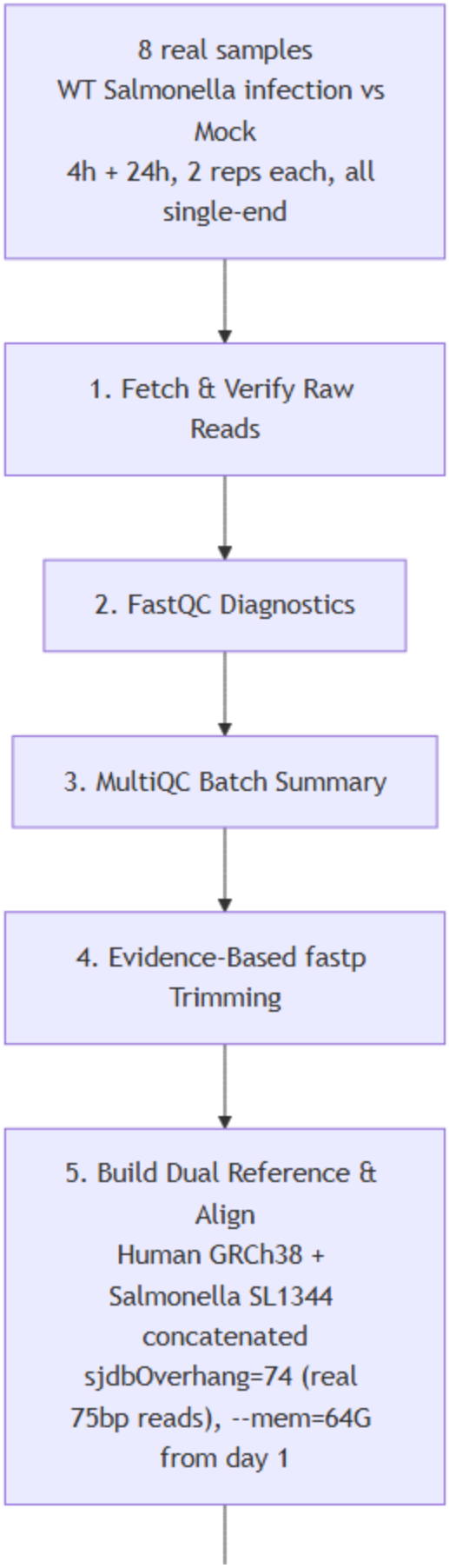
4.1 Task 1 (HPC)

- Slurm account, /scratch allocation, internet egress from compute nodes.
- Modules: SRA-Toolkit, FastQC, fastp, STAR, SAMtools, Subread, MultiQC.
- Memory: 64 GB for the combined human+Salmonella STAR index — applied FROM THE START, not discovered the hard way (see the companion PRJNA1091619 pipeline's own real OOM history).

4.2 Task 2 (R)

- R (recent 4.x) with Bioconductor access; RStudio recommended.
- Packages: ggplot2, dplyr, tidyr, tibble, pheatmap, RColorBrewer, patchwork, ggrepel, data.table, DESeq2.

5. Task 1 — Raw Reads to Counts (HPC Bash Pipeline)


8 real samples
WT Salmonella infection vs Mock
4h + 24h, 2 reps each, all single-end
1. Fetch & Verify Raw Reads
2. FastQC Diagnostics
3. MultiQC Batch Summary
4. Evidence-Based fastp Trimming
5. Build Dual Reference & Align
Human GRCh38 + Salmonella SL1344 concatenated
sjdbOverhang=74 (real 75bp reads), --mem=64G from day 1

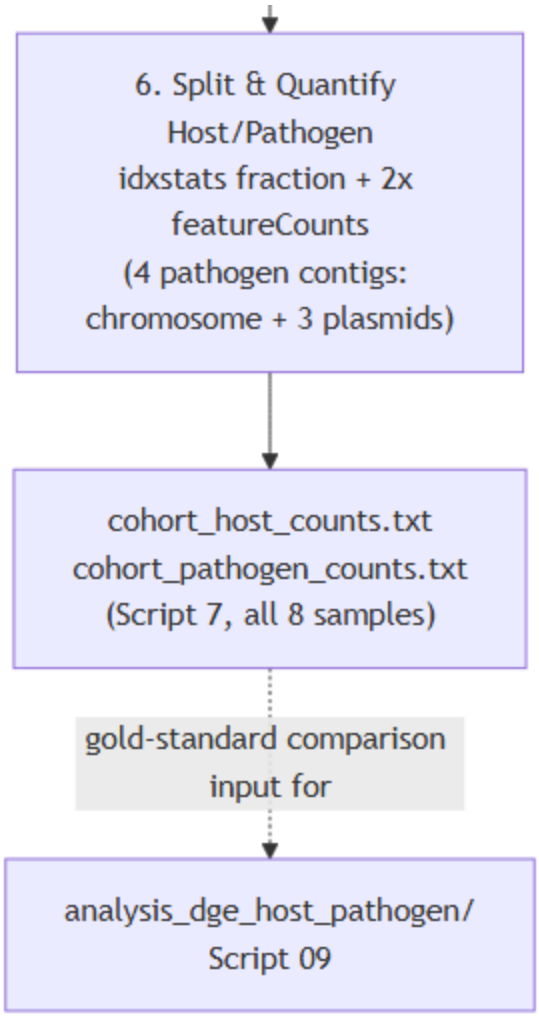


Figure 1. Raw-reads-to-counts pipeline for the 8-sample representative subset.

Step 1 — Fetch & Verify Raw Reads

```
sbatch 1_Fetch_And_Verify_Raw_Reads.sh
```

Step 2 — FastQC Diagnostics

```
sbatch 2_Interpret_FastQC_Diagnostics.sh
```

Step 3 — MultiQC Batch Summary

```
sbatch 3_Batch_MultiQC_Summary.sh
```

Step 4 — Evidence-Based fastp Trimming

```
sbatch 4_Evidence_Based_Fastp_Trimming.sh
```

Step 5 — Build Dual Reference & Align

```
sbatch 5_Build_Dual_Reference_And_Align.sh
```

- sjdbOverhang 74 = this study's own real 75bp read length - 1 (verified via SRA run metadata).
- --mem=64G and genomeParameters.txt-based idempotency from the start — real lessons already learned on the companion PRJNA1091619 pipeline, applied proactively here.

Step 6 — Split & Quantify Host/Pathogen

```
sbatch 6_Split_And_Quantify_Host_Pathogen.sh
```

Four pathogen contigs, not one

Salmonella SL1344's reference has a main chromosome (NC_016810.1) PLUS three plasmids (NC_017718/019/720.1) — all four must be treated as "pathogen" in the host/pathogen read-fraction diagnostic, unlike the single-contig Mtb pipeline.

Step 7 — Automated Cohort Run (all 8 samples)

```
sbatch 7_Automated_Pipeline_AllSamples.sh
```

Step 8 — QC Report Generator

```
bash 8_QC_Report_Generator.sh   # run locally, NOT via sbatch
```

6. Task 2 — DESeq2 + ggplot2 Analysis (R)

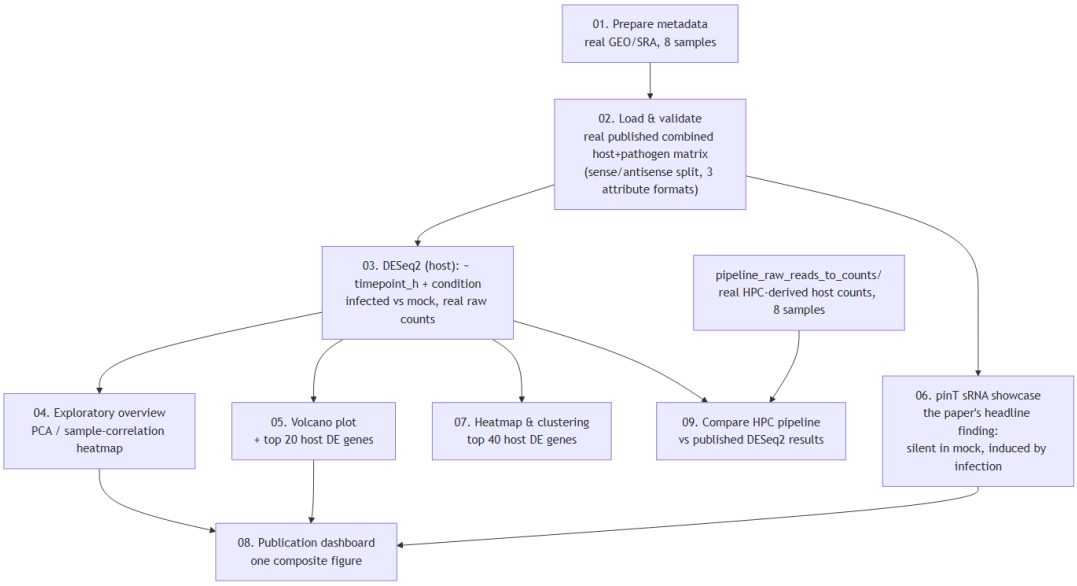


Figure 2. The analysis workflow, run on the real published combined count matrix.

Step 00-02 — Setup, Metadata, Data Loading

```
source("00_install_packages.R"); manage_packages("automatic")
source("01_prepare_metadata.R")
source("02_load_and_validate_counts.R")
```

Read Script 02's own comments before trusting its output

The published file is tab-delimited despite its .csv extension, has separate sense/antisense rows per feature (only sense is kept), and uses THREE different attribute-encoding conventions across host genes, pathogen CDS features, and pathogen sRNA features — an extraction pattern built for only one of these would silently miss real genes, including pinT itself.

Step 03 — DESeq2 (host, the primary analysis)

```
source("03_differential_expression_deseq2.R")
```

- Design: ~ timepoint_h + condition — controls for the real 4h/24h timepoint covariate.

Step 04 — Exploratory Overview

```
source("04_exploratory_overview.R")
```

Step 05 — Volcano Plot

```
source("05_volcano_and_top_genes.R")
```

Step 06 — pinT sRNA Showcase

```
source("06_pathogen_pint_showcase.R")
```

- Real, independently-recomputed result: pinT counts are 0/0/0/2 across all mock samples, but 191.5 mean at 4h infection and 1,837.5 mean at 24h infection.

Step 07 — Heatmap & Clustering

```
source("07_heatmap_and_clustering.R")
```

Step 08 — Publication Dashboard

```
source("08_publication_dashboard.R")
```

Step 09 — Compare HPC Pipeline vs Published Results

```
source("09_compare_hpc_pipeline_to_published.R")
```

- Corrects a real ID-format mismatch: published gene IDs carry a GENCODE version suffix (ENSG...NN), this pipeline's own Ensembl GTF does not — stripped explicitly before joining, not silently ignored.

7. Downstream Handoff Between the Two Tasks

```
# On your local machine / RStudio project root, after Task 1's Script 7 completes:
mkdir -p pipeline_raw_reads_to_counts_output/counts
scp <cluster>:/scratch/<you>/PRJNA280839_salmonella_automated/counts/cohort_host_counts.txt* \
    pipeline_raw_reads_to_counts_output/counts/
```

Appendix A — Package Reference

Package	Source	Purpose
ggplot2	CRAN	General plotting
dplyr / tidyr / tibble	CRAN	Data manipulation and reshaping
data.table	CRAN	Fast loading of the large published count file
pheatmap / RColorBrewer	CRAN	Heatmaps and palettes
patchwork	CRAN	Composite multi-panel figures
ggrepel	CRAN	Non-overlapping plot labels
DESeq2	Bioconductor	Primary differential expression analysis

Appendix B — Resource Allocation Reference

Script	CPUs	Memory	Wall time
1_Fetch_And_Verify_Raw_Reads.sh	4	8 GB	1 h
2_Interpret_FastQC_Diagnostics.sh	4	6 GB	45 min
3_Batch_MultiQC_Summary.sh	4	8 GB	1.5 h
4_Evidence_Based_Fastp_Trimming.sh	4	8 GB	1 h
5_Build_Dual_Reference_And_Align.sh	8	64 GB	3 h
6_Split_And_Quantify_Host_Pathogen.sh	4	8 GB	45 min
7_Automated_Pipeline_AllSamples.sh	8	64 GB	12 h
8_QC_Report_Generator.sh	1 (local)	4 GB	10 min

Appendix C — Data Provenance Table

Item	Value
GEO SuperSeries / sub-series	GSE60144 / GSE67758 (51 samples total; this project uses 8)
BioProject	PRJNA280839
Publication	Westermann et al. 2016, Nature 529(7587):496-501, doi:10.1038/nature16547
Host reference	Ensembl GRCh38, release 110
Pathogen reference	NCBI RefSeq GCF_000210855.2 (ASM21085v2)
Headline gene	pinT, locus_tag ncRNA0164, chromosome NC_016810.1
Published count matrix	GSE67758_Raw_gene_wise_quantifications_Human_and_Salmonella.csv.gz

Citation

Westermann AJ, Förstner KU, Amman F, Barquist L, Chao Y, Schulte LN, Müller L, Reinhardt R, Stadler PF, Vogel J. (2016) Dual RNA-seq unveils noncoding RNA functions in host-pathogen interactions. *Nature* 529(7587):496-501. https://doi.org/10.1038/nature16547